Quantitative Analysis of Vertical Motion of a Bouncing Ball Using Video Tracking

Introduction:

The motion of objects under gravity is a fundamental concept in physics. When an object is projected vertically, it follows a parabolic path due to constant acceleration caused by gravity.

In this experiment, video tracking techniques were used to analyze the vertical motion of a bouncing ball. By extracting position-time data and fitting a quadratic model, the nature of the motion and the consistency of gravitational acceleration were investigated.

Objective:

To analyze the vertical motion of a bouncing ball and determine whether it follows a quadratic relationship with time, and to estimate the acceleration due to gravity using video tracking.

Apparatus:

- Mobile phone camera
- Kinovea software
- Microsoft Excel
- Bouncing ball
- Measuring scale (for reference)

Method:

1. A video of a bouncing ball was recorded using a mobile phone camera.

2. The motion of the ball was analyzed using Kinovea by placing a marker at the center of the ball frame-by-frame.

3. The vertical position (Y) and time (t) data were extracted and exported to Microsoft Excel.

4. The data was cleaned by selecting only the frames corresponding to a single bounce to ensure a smooth parabolic motion.

5. Separate scatter graphs of Y against time were plotted for two different bounces.

6. Polynomial trendlines of order 2 were fitted to each graph.

7. The equations and R^2 values were obtained from the graphs.

Graphs:

Note: Green markers () represent the vertical position of (Y) of the ball.

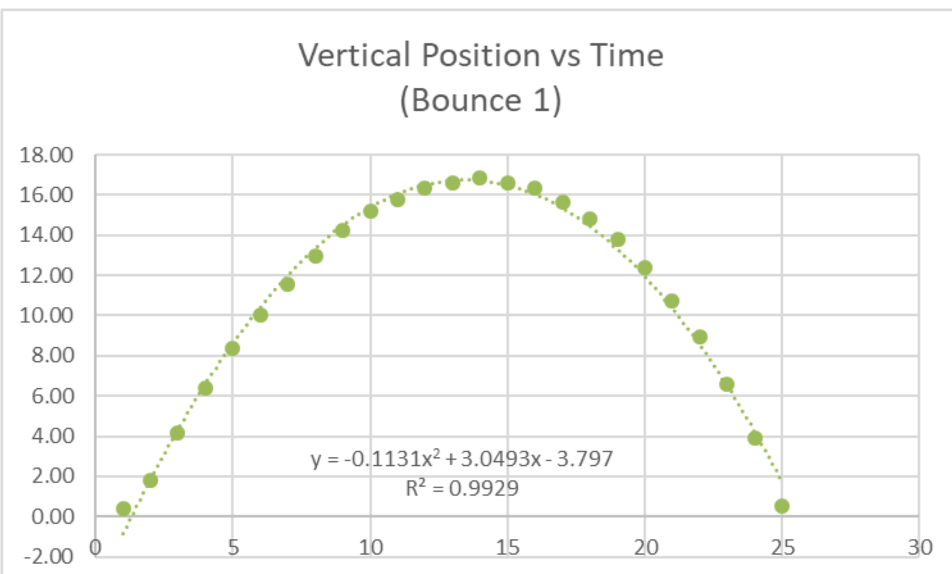

Figure 1: Vertical position vs time for Bounce 1

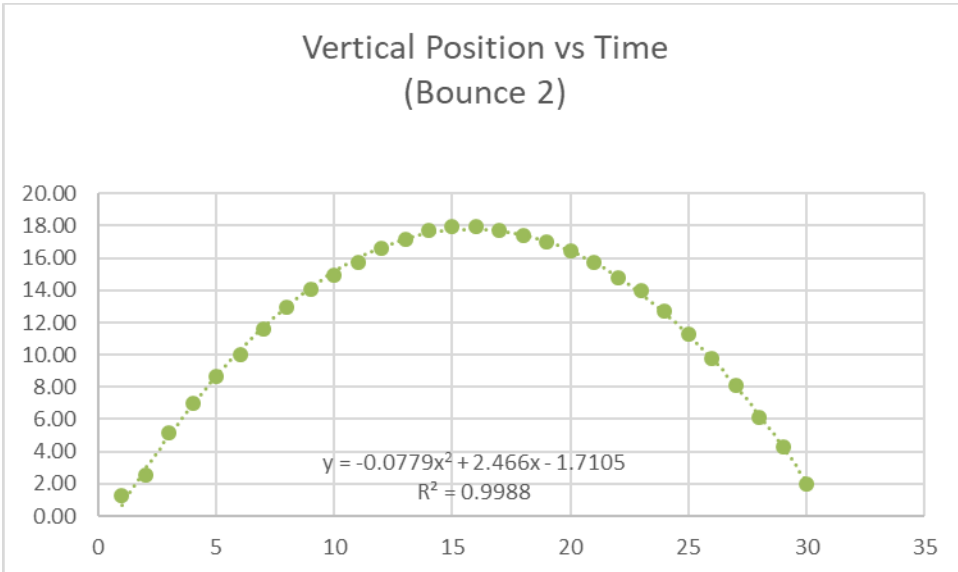

Figure 2: Vertical position vs time for Bounce 2

Results:

The motion of the ball was found to follow a quadratic equation of the form:

$Y = ax^2 + bx + c$

where Y is the vertical position (cm) and x is time (s).

Bounce 1:

$Y = -0.1131x^2 + 3.0493x - 3.797$

$R^2 = 0.9929$

Bounce 2:

$Y = -0.0779x^2 + 2.466x - 1.7105$

$R^2 = 0.9988$

Calculation of Acceleration Due to Gravity:

The general equation of motion under constant acceleration is:

$Y = (1/2)gt^2 + v_0t + Y_0$

Comparing with:

$Y = ax^2 + bx + c$

we get:

$a = (1/2)g$

Therefore:

$g = 2a$

Bounce 1:

$a = -0.1131$

$g = 2 \times 0.1131 = 0.2262\ cm/s^2$

$g = 0.002262\ m/s^2$

Bounce 2:

$a = -0.0779$

$g = 2 \times 0.0779 = 0.1558\ cm/s^2$

$g = 0.001558\ m/s^2$

Comparison:

Bounce	Coefficient (a)	R^2 Value	g (cm/s^2)	g (m/s^2)
1	-0.1131	0.9929	0.2262	0.002262
2	-0.0779	0.9988	0.1558	0.001558

(The results obtained from both bounces are summarized in the table above)

Analysis:

The motion of the ball in both cases closely follows a quadratic relationship, confirming that the motion is uniformly accelerated under gravity.

The high R^2 values (0.9929 and 0.9988) indicate an excellent fit of the experimental data to the theoretical model. This demonstrates that the tracking and data processing methods were effective.

The calculated values of acceleration due to gravity are lower than the theoretical value of 9.8 m/s^2. This difference is primarily due to scaling limitations in time measurement and possible perspective errors from the camera.

Despite this, the consistency between the two bounces suggests that the experimental method is reliable and repeatable.

Evaluation:

Several factors may have affected the accuracy of the results. These include minor errors in marker placement during video tracking, perspective distortion due to camera angle, and limitations in frame rate affecting time measurement.

Improvements could include using a higher frame rate camera, ensuring a perpendicular camera angle, and increasing the number of trials to improve reliability

Conclusion:

This experiment successfully demonstrated that the vertical motion of a bouncing ball follows a quadratic relationship with time, confirming uniformly accelerated motion under gravity.

The high R^2 values obtained for both bounces (0.9929 and 0.9988) indicate excellent agreement between the experimental data and the theoretical model.

Although the calculated values of gravitational acceleration differ from the standard value, this is likely due to limitations in time scaling and experimental setup.

Overall, the results are consistent and reliable, and the use of video tracking proved to be an effective method for analyzing motion and extracting meaningful physical quantities.